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10 CONCEPTUAL Q&A PROGRAMMING LANGUAGES · PRACTICE WORKSHEET

CONCEPTUAL SERIES
C++

Name: ______ Cohort/Batch: ______ Date: ______ Score: ______

C++ PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Programming Languages

TOTAL: 10 QUESTIONS

Q1 What are the primary design goals of C++? — Threat Model

Q2 How does C++ handle type checking: static or dynamic? — Architecture

Q3 How does C++ manage memory? — Configuration

Q4 What is C++'s concurrency model? — Hardening

Q5 How are packages and dependencies managed in C++? — SSO / IdP

Q6 How does C++ implement object-oriented or functional programming? — Observability

Q7 How does C++ handle errors? — Lifecycle

Q8 What testing tools are commonly used in C++? — Compliance

Q9 What makes C++ well-suited for its target domain? — Testing

Q10 What are common performance pitfalls in C++? — Tuning

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 C++ was designed with specific priorities
Mitigation

C++ was designed with specific priorities around performance, safety, or developer productivity depending on its domain. These goals shape its syntax, type system, and standard library choices.

A2 C++ uses its type system to
Primitives

C++ uses its type system to catch errors at compile time (static) or at runtime (dynamic). This affects refactoring safety, tooling support, and the kinds of bugs that reach production.

A3 C++ uses one of three approaches:
Hardening

C++ uses one of three approaches: manual allocation/deallocation, a garbage collector that periodically reclaims unreachable objects, or automatic reference counting. Each has different latency and throughput tradeoffs.

A4 C++ supports concurrency through threads,
Gotchas

C++ supports concurrency through threads, coroutines, async/await, or an actor model. The choice determines how I/O-bound and CPU-bound workloads are scaled and how shared state is safely accessed.

A5 C++ uses a package manager and
Federation

C++ uses a package manager and a manifest file to declare dependencies and version constraints. A lock file pins exact versions to ensure reproducible builds across environments.

A6 C++ supports classes and inheritance for
Observability

C++ supports classes and inheritance for OOP, or first-class functions and immutability for FP, or both. Many modern C++ programs blend both paradigms depending on the problem.

A7 C++ surfaces errors via exceptions, Result/Either
Automation

C++ surfaces errors via exceptions, Result/Either types, or error return values. The choice impacts whether errors are checked at compile time and how calling code is structured.

A8 C++ has a standard or widely
Governance

C++ has a standard or widely adopted test runner and assertion library. Tests are typically organized into unit, integration, and end-to-end layers with coverage reporting built in or via plugins.

A9 C++ excels in its domain due
Assessment

C++ excels in its domain due to a combination of performance characteristics, ecosystem maturity, language ergonomics, and community support that makes common tasks simple.

A10 Typical performance issues in C++ include
Optimization

Typical performance issues in C++ include excessive allocations, blocking the main thread or event loop, O(n^2) data structures, and missing caching. Profiling and benchmarking tools are available to diagnose them.